Human-Centered Design for Care Coordination and Adherence in the ICU

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Abstract

Adherence to standards of care remains an ongoing healthcare challenge. We explored this challenge with adherence to Wake Up and Breathe in the ICU. Previous work suggested a dashboard could improve coordination between clinicians, a notable challenge to adherence. To explore this problem space, we conducted co-design sessions with ICU clinicians. We found clinicians did not view dashboard can change their behavior and coordination as problematic, however, concerned about documentation burden. Drifting by intention, we pivoted toward solutions that address both clinicians' needs and patient outcomes. During a sketching session, we envisioned an auto-documentation concept designed to capture WUB actions and make their coordination visible in real-time. In enactment sessions, participants expressed acceptance of the concept and articulated its value. Our findings identify a design opportunity that highlights how auto-documentation might improve adherence by generating real-time ICU data that improves care coordination, and we discuss broader design implications beyond the ICU.

Keywords

Healthcare, Co-design, Auto-documentation, Care Coordination, Standards of care, Adherence, ICU

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1 Introduction

Healthcare employs clinical trials and meta-analyses to produce evidence-based standards of care [34, 41, 67, 86]. These standards help maximize patient outcomes. However, implementation in noisy, chaotic clinical environments can be challenging and often leads to low adherence rates [79]. HCI researchers have begun exploring how socio-technical interventions might bridge this gap by designing digital systems that support clinical team coordination and communication during critical care delivery [9, 87].

Recently, HCI researchers investigated the 50% adherence rate for the Wake Up and Breathe protocol (WUB) in the intensive care unit (ICU) setting [99]. WUB is the standard of care for assessing when patients should be liberated from mechanical ventilators. Adherence to WUB substantially reduces mortality and other adverse outcomes [50]. WUB requires coordination and collaboration between nurses and respiratory therapists (RTs). Nurses cut the patient's sedation. After the patient wakes up, RTs cut breathing support to assess whether the patient can breathe on their own. Prior HCI research suggested that dashboards displaying a WUB schedule could reduce ambiguity over which patients qualify for WUB, improve coordination between nurses and RTs, and increase adherence [98, 99].

To further explore the solution space, we conducted two co-design experiments with ICU nurses and RTs who regularly perform WUB. In our Design Experiment 1, we probed the idea of

shared dashboards as a solution to the adherence challenge. Surprisingly, participants questioned whether a shared dashboard would change their behavior or improve care coordination. They repeatedly pushed the co-design focus towards the burnout they experienced from their documentation work. In a Design Experiment 2, we combined their desire for less documentation and our interest in adherence to standards of care. We first conceived of an auto-documentation concept that monitors and documents the execution of WUB. We then engaged ICU clinicians in co-design, asking them to enact their vision of this concept. Participants indicated that they would accept and use this kind of concept, and that they would strongly prefer it over their current work process if it reduced manual charting. Reflecting on the auto-documentation concept, we found that it offers potential benefits for clinicians, patients, healthcare systems, and technical innovators: it would improve adherence by generating real-time ICU data to support better care coordination.

Our research makes three contributions. First, it demonstrates the value of *Drifting by Intention* in co-design for high-stake ICU. Second, it proposes auto-documentation of standards of care as one way to simultaneously address the challenges of adherence and clinician burnout. Third, it provides a reminder that HCI interventions work best when they address worker and enterprise needs simultaneously. This challenges the unarticulated convention that today's healthcare IT systems seem to ignore clinician needs, leading to problems like alert fatigue. While we focused on WUB in the ICU, we see potential for auto-documentation of standards in healthcare to help with adherence more broadly.

2 Background and Related Work

Below, we provide background on the ICU and WUB protocol and computing in clinical practice. We then review HCI literature on how computing advances support team coordination. Lastly, we review HCI literature on the ICU.

2.1 Background on the ICU and Wake Up and Breathe Protocol

Intensive care units (ICUs) provide care for the sickest patients in a high-stress, dynamic, chaotic, fast-paced environment [21, 91]. One life-saving intervention frequently used in the ICU is mechanical ventilation, used in approximately 20% of ICU admissions, representing approximately 1.6 million patients annually in the United States [37]. Ventilators take over the work of breathing so a patient can rest and heal [40]. Clinicians intubate patients under sedation so they can tolerate a tube in their airway and a machine that pushes air into and pulls air out of their lungs. ICU patients on ventilators are critically ill. Their mortality rate of approximately 35-50% is significantly higher than the 10-29% mortality rate for all ICU patients [56].

Prolonged ventilation increases the risk of harm and death [62, 71]. Best practices for ventilator management aim to reduce total ventilation time, minimize sedation, and limit the support a ventilator provides [30]. The Wake Up and Breathe (WUB) protocol promotes safe and timely extubation by guiding clinicians in deciding when to liberate patients from ventilation [38]. The WUB protocol should be performed once each day. It includes two key components: Spontaneous Awakening Trials (SATs), conducted by nurses, and Spontaneous Breathing Trials (SBTs), executed by respiratory therapists (RTs). Fig 1 below provides an overview of how nurses and RTs coordinate. WUB begins with a SAT, where nurses cut sedation to awaken the patient and assess cognitive function and tolerance to discomfort. Next, RTs conduct SBT, removing ventilator support, and monitor whether the patient can successfully breathe on their own.

Effective execution of WUB requires collaboration and coordination between nurses and RTs. In reality, they have quite different collaboration experiences [12, 94]. ICU nurses typically care for two patients at a time, though rarely is more than one patient ventilated simultaneously. In contrast, a single RT cares for all patients on breathing support within a single ICU. Most nurses work a 12-hour shift, while RTs work 8-hour shifts. Nurses and RTs start and end their days at different times, and when they gather at the start of a shift, they meet only with peers in their same role. The RTs need to arrive at a patient's room soon after the nurse has started SAT. If the RT arrives too early, they must wait for the patient to wake up and tolerate being awake. if they arrive too late, the patient is back on sedation, leading to no SBT or an SBT given to a fully sedated patient [33, 43]. A nurse should not start an SAT if another nurse in the same ICU has already started SAT. Currently, nurses and RTs have no access to real-time data detailing what other clinicians in the ICU are doing or plan to do, in order to collaborate and coordinate more effectively [83, 92].

Despite the clear benefits of WUB, less than 50% of patients who qualify for WUB receive an SAT and SBT [2, 55]. Researchers have investigated the barriers and facilitators of the WUB protocol to support its effective execution [2, 6, 8, 20, 23, 44, 85]. This body of work highlights a number of clinical and behavioral barriers, including safety concerns such as fear of patient agitation or accidental self-extubation [2, 85], mistaken beliefs that sedation helps patients rest and heal [24, 36], and high workload with many competing priorities [6]. Furthermore, prior research shows that ICUs with better coordination and collaboration have better adherence to standards of care [8, 23].

2.2 Computing in Clinical Practice

HCI has a long history of exploring how computing might transform clinical practice and improve patient outcomes. For example, researchers are exploring technical opportunities, such as clinical decision support systems [26, 51, 78], which leverage AI to enhance clinicians' diagnostic and treatment decisions. Researchers have also developed AI systems that predict the need for mechanical ventilation [84] and estimate the risk of extubation failure [89]. Moreover, HCI researchers have explored ways to integrate computing systems into clinicians' practice [97] by examining how clinicians engage with and adopt electronic health records (EHR) [59, 60].

Unfortunately, IT systems deployed in clinical practice are often not experienced by clinicians as beneficial. Many address problems that are not particularly important to clinicians [74], or they have been designed with little concern for the realities of clinical practice. For example, many monitoring systems attempt to add value by issuing alerts for potential problems a clinician might overlook.

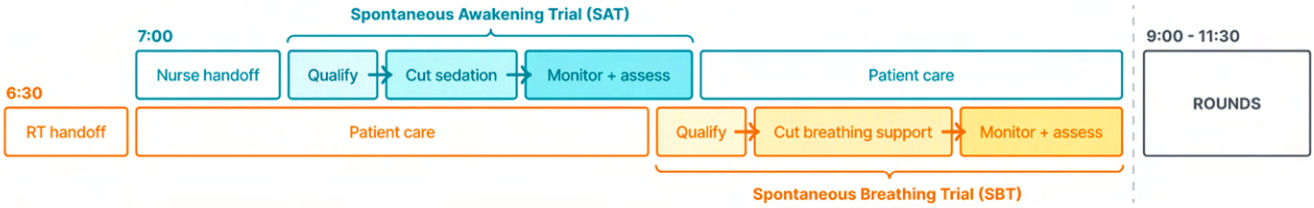


Figure 1: This diagram illustrates how nurses and RTs coordinate the performance of the nurse's Spontaneous Awakening Trial (SAT, green) and the Respiratory Therapist's Spontaneous Breathing Trial (SBT, orange) as part of the Wake Up and Breathe protocol [99].

However, these systems often generate *alert fatigue* [53, 64, 68]. An ICU nurse might experience more than a thousand alerts during a single shift [73]. Clinicians' alert fatigue makes them quit attending to alerts. Similarly, the introduction of EHR, meant to improve patient care, has created a huge documentation burden [4]. These systems come between clinicians and patients during their interactions. The heavy documentation burden forces clinicians to work beyond their shift hours, completing tasks designed for hospital accountability rather than supporting individual clinical workflows [10, 80].

To mitigate this documentation burden, researchers have developed digital scribes [16, 70]. While commercial tools are increasingly adopted, most are limited to patient-clinician consultations [1, 75]. This remains limited in fast-paced environments such as the ICU or Emergency Department. To address this gap, recent work has incorporated multimodal sensing [27, 47]. However, these systems introduce severe privacy and surveillance concerns and critical questions remain regarding clinician acceptance through a human-centered lens.

2.3 HCI Advances in Team Coordination

HCI research has investigated how computing can improve coordination across teams in healthcare [3], software engineering [11], design [49], and the remote workplace [66, 93]. One common strategy involves the use of dashboards that display what team members are doing, what has been completed, and what needs to be done [25]. These dashboards can influence group behavior [25], foster shared situational awareness [13], and provide at-a-glance visibility into team activities [61]. For example, restaurants use shared displays to coordinate between the kitchen workers and frontline workers. Orders are entered, tracked, and fulfilled in real time, which improves efficiency, reduces errors, and enhances the customer experience. Shared displays have been shown to prevent conflicts in collaborative environments. In software development, FASTDash visualized which files each programmer is working on, improving awareness and reducing conflict [11]. In design settings, interactive whiteboards supported more effective collaboration among designers [49].

Improving coordination in healthcare presents unique challenges for HCI. Patients receive treatment from multiple providers with diverse roles, requiring complex collaboration. In the ICU, all team members must stay aligned on who did what and when. Breakdowns in awareness can compromise patient safety [65]. Enhanced clinical care coordination not only enhances patient outcomes, but also reduces waste and delays, leading to more efficient care at a lower cost [57]. Building on HCI strategies to support coordination, research in clinical settings has explored how computing can enable seamless workflows. For example, digital checklists have been used to coordinate care during heart attack interventions, reducing human error and improving communication and performance at critical interventions [5, 88]. Similarly, Wong et al. explored the implementation of an electronic whiteboard in General Internal Medicine [96], finding that it helped standardize communication within the care team by providing a snapshot view of patient activity.

2.4 HCI Research on the ICU

The complex, fast-paced environment of the ICU has increasingly drawn the attention of the HCI research community. Studies have explored the challenges faced by ICU clinicians through field studies, surveys, interviews, and meta-syntheses [28, 54, 63, 76, 95, 98, 99]. Other research has investigated ways to support ICU care teams by designing and evaluating AI systems, for example, by examining how clinicians interact with real AI-based treatment recommendations [76] and exploring how to communicate prediction uncertainty in ICU [54]. Researchers have also investigated ICU clinician workflows to understand how to integrate computing systems into their clinical practices, focusing on goals such as progress note documentation [95] and WUB adherence [98, 99].

In particular, research by Yildirim et al. conducted co-design studies with ICU nurses and RTs to examine adherence to the WUB protocol [98, 99]. They identified adherence barriers, including ambiguity over which patients qualify for WUB and difficulty in prioritizing WUB patients. They discovered that when busy clinicians are faced with uncertainty, such as how to coordinate WUB, they ignore uncertain tasks and focus on other work. Based on these findings, the authors proposed that the shared dashboard displaying a schedule for WUB could reduce ambiguity, improve coordination, and increase adherence. They assumed that the ICUs with higher levels of coordination with a dashboard have better adherence to the WUB and have better healthcare outcomes, including fewer days on a ventilator, shorter stays in ICU, and fewer deaths.

Building on this line of research, our paper takes the dashboard concept as a starting point and aims to explore several open questions the prior work left: Would clinicians accept such a dashboard in real-world clinical practice? When and where might it be situated in their workflow? And what information would it need to display to be effective?

3 Overview of the Design Process

We explored the idea that a dashboard could improve WUB adherence by improving coordination between RTs and Nurses [99]. We chose to use a Research through Design (RtD) approach, which leverages aspects of design practice to explore and reframe problematic situations [31, 72, 100]. RtD projects employ *design experiments*, the intentional actions (design moves) taken by design researchers to move a project forward [100, 101]. We conducted two design experiments, both using co-design, to envision a socio-technical system that can increase clinical adherence.

Co-design is a participatory approach in which stakeholders work alongside researchers as active design contributors [42, 52, 81]. Rather than directly asking users for solutions, co-design engages users in activities that reveal their needs, values, desires, and fears [69]. This approach offers two key benefits for our study: 1) Surfaces friction and problematic dependencies across roles [82], making it well-suited to address tensions between the nurses and RTs in our study. 2) Reduces the risk of non-adoption and increases worker acceptance and use by including stakeholders in the design [39, 77]. We conducted co-design activity grounded in Brandt et al. [17]'s framework of *tell*, *make*, and *enact*, in which participants share stories about their current work, design future ways of working, and enact new work practices.

Design Experiment 1 explored how clinicians might engage with a dashboard designed to improve WUB protocol adherence and standards of care. We conducted eight co-design sessions with nurses and RTs who perform the WUB protocol. These drove us to pivot toward an emergent solution space that simultaneously addresses healthcare and clinician needs. Design Experiment 2 involved sketching concepts that address burnout and adherence. We then ran three co-design sessions with nurses and RTs to act out what their work might be like with an auto-documentation system and assess if this might be acceptable. Every Design Experiments were approved by our institution's IRB.

4 Design Experiment 1: Improving WUB Adherence with Dashboards

4.1 DE1 Process

We began the workshop by acknowledging the significant contributions of ICU nurses and RTs within the chaotic, high-stakes ICU (Fig 2). We wanted to avoid participants feeling defensive when discussing the lack of adherence, and to show our team's empathy for their work challenges in the face of their dedication and professionalism. We then moved on to a co-design activity grounded in Brandt et al. [17]'s framework.

4.1.1 Prompting with Dashboards. We first presented an unfinished dashboard displaying a schedule for WUB (ranked by patients most likely to be extubated) [98, 99] (Fig 3). The incomplete dashboard

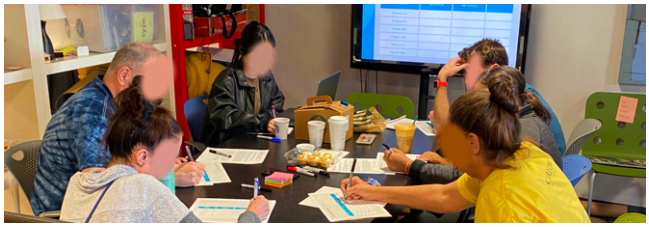


Figure 2: Study environment of DE1's co-design sessions showing how participants engaged in the study.

was intended as a starting point for co-design [77, 90]. This led us to probe several open questions: Would clinicians accept and use such a dashboard, When and where might it be situated in their workflow, and what information would it need to display to be effective? We encouraged participants to share their current experiences, envision future ways of coordinating, and refine, propose, discard, or redirect ideas, supporting them to consider a broader range of potential solutions beyond the dashboard itself.

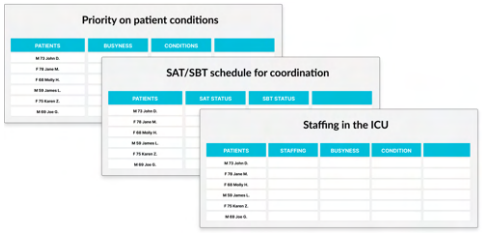


Figure 3: Sketch of a Shared Dashboard used during co-design studies to probe participants on how a schedule for WUB based on patients' likelihood of being liberated from a ventilator might help improve collaboration between nurses and RTs.

4.1.2 Reframing. We analyzed the collected data after each workshop to inform and adapt the design of subsequent sessions. These moments of reflection allowed us to monitor whether the project's design intentions were shifting and to adapt subsequent activities accordingly. Prior design research has shown that Research through Design often proceeds through design experiments that generate new research questions by reframing the problem [45]. This process also known as *Drifting by Intention*, is an integral part of the design process.

4.1.3 Envisioning with Scenarios. Reframing shifted the focus of the workshop to investigating expectations about what good coordination looks like. We developed new scenarios around the coordination barriers that clinicians surfaced in the first six workshops (Fig 4). We again deliberately left these scenarios incomplete, using them as *make* triggers to help clinicians envision a beneficial and acceptable future for their work rather than to prescribe one.

We designed prompts and scenarios to challenge clinicians' existing thinking about what good coordination should look like [19, 48]. We introduced an example from a non-clinical setting: the collaboration between cashiers and baristas at Starbucks, who coordinate

seamlessly to quickly and accurately produce the customized drink orders. This analogy helped participants think outside the box and prompted reflection and discussion about their own practices, encouraging them to consider whether they currently have or should aim to achieve a similar quality of coordination in their execution of WUB.

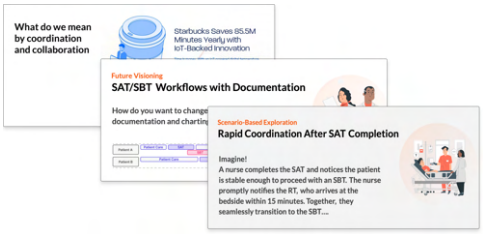


Figure 4: Materials used in the last two co-design sessions to probe participants through scenario-based exploration.

4.1.4 Participants. We recruited 14 nurses and 17 RTs (N=31) from three hospitals to participate in the study. They had a mean of approximately fifteen years of ICU work experience. All had worked in the ICU for more than a year, and eighteen for more than ten years. Each workshop included three to nine participants, and each had at least one nurse and one RT. They lasted between one and a half to two hours. We paid each participant $150 for their valuable time. Table 1 provides a summary of participants' roles, years of experience, and the workshop they participated in. Note: The final two workshops reused participants from the first six workshops.

4.1.5 Data Analysis. The data collected during the workshops include video recordings, transcripts produced by Otter.AI, photos of participant-made artifacts (Fig 5), and notes taken by the research team. The research team viewed the videos, updating the transcripts with details describing participants' facial expressions, emotions, reactions, and gestures. We analyzed all of this qualitative data by creating affinity diagrams [35] with the collaboration software Miro. We individually reviewed the data, capturing key insights as Miro sticky notes. Next, we held meetings to cluster the notes and to discuss and refine emerging clusters.

4.2 DE1 Findings

Across all workshops, RTs and nurses maintained a positive, friendly, and productive atmosphere. Most shared that they enjoyed the co-design experience. They seemed happy to participate and were authentically interested in helping create new healthcare computing systems. Some nurses and RTs had previously worked together, offering them a shared set of work experiences to start from. Participants said they enjoyed sharing stories about their work, hearing about practices in other units, and listening to others' stories. A few participants seemed cautious to share at first; however, this hesitance dissipated quickly. We observed participants trying to show mutual respect, and many highlighted the positive aspects of clinicians from the other role.

The analysis produced three high-level findings:

(1) RTs and nurses both agreed that a dashboard was unlikely to change the way they work or increase adherence.

Table 1: Participant Demographics for DE1

Participant Code	Role	Gender	Years of Experience
N1_1	Nurse	F	16
R1_1	RT	F	-
R1_2	RT	M	20
R2_1	RT	M	20
R2_2	RT	M	31
N2_1	Nurse	F	2.5
N3_1	Nurse	M	1.5
R3_1	RT	M	18
R3_2	RT	F	16
R3_3	RT	F	20
R3_4	RT	F	17
N4_1	Nurse	F	26
R4_1	RT	F	10
N4_2	Nurse	F	7
R4_2	RT	F	22
R4_3	RT	M	12
N4_3	Nurse	M	1.5
R5_1	RT	M	-
N5_1	Nurse	M	2.5
N5_2	Nurse	F	3
R5_2	RT	F	7
N5_3	Nurse	F	1.5
R5_3	RT	M	33
N5_4	Nurse	M	3
R5_4	RT	F	20
N5_5	Nurse	F	37
R6_1	RT	M	15
N6_1	Nurse	F	1
R6_2	RT	M	35
N6_2	Nurse	F	5
N6_3	Nurse	F	28
N7_1	Nurse	F	28
R7_1	RT	M	15
N7_2	Nurse	M	2.5
R7_2	RT	M	12
N8_1	Nurse	F	5
N8_2	Nurse	F	1
R8_1	RT	F	7
R8_2	RT	F	22
N8_3	Nurse	F	26

Note. "–" indicates that participants chose not to answer. The final two workshops reused participants from the first six workshops.

(2) RTs and nurses both felt they already did an excellent job of coordinating, viewing communication as synonymous with coordination.
(3) RTs and nurses both asked for systems that would reduce their documentation burden and their experience of burnout.

4.2.1 ***Negative View of Screens and Dashboards***. During the first few workshops, we provided participants with an incomplete dashboard. We wanted them to think about where this might go in the ICU and the information they might need to make use of it. Participants were highly skeptical of a shared dashboard. When asked to complete or extend the dashboard, many participants made only half-hearted attempts. They questioned whether the insights from the prior dashboard [99] would transfer to their workflow. Their concerns centered on limited time, rapidly changing patient status, and the absence of timely data.

Participants imagined the dashboard as a screen they would navigate to within the EHR, rather than as a persistent display. As R3_1 shared, "*We are covering too much area, so I won't look at dashboard ... So something that's in the computer doesn't necessarily work.*" Similarly, R3_3 noted, "*So there's really not a good way to see it in a charting system or in a shared dashboard, because I'm going to miss my SBT. You have to, as soon as they're awake, you have to put them on; you may miss your window. So to see it in a shared dashboard is not a good way of relying on that information. Your best bet is real-time communication from your nurse.*" They viewed the computer as a device that removed them from caring for patients, not as a tool that might make them better at their job. Moreover, given experience of coordinating care without screens, they found it difficult to consider screens playing any role.

Some participants identified specific situations where a shared dashboard might help. For example, N6_3 reflected: "*I think the shared dashboard is a good start... It's a good start to your day. But I mean, if things go south, then obviously the shared dashboard would not work... A potential problem with shared dashboards is that patient status changes so quickly, and the shared dashboard won't be up to date because charting doesn't happen immediately.*" They recognized that a dashboard without real-time data was a waste of time.

Participants struggled with the idea of a shared dashboard. Their experience of always working one-on-one with a computer conflicted with the notion of a shared digital display. Instead, they suggested individual uses, such as a "*to do*" list, rather than coordinating actions between nurses and RTs. N5_2 explained: "*It [our concept of a shared dashboard] will keep just like the task list. You know, like the nurses' task list reminds them, did you do all these tasks? So, like, I think the task list, even though sometimes they hate it, I think it's not a bad thing, it's a reminder, because then, you know, you're good.*"

In general, RTs and nurses viewed the effort in using a computer as greater than the benefit. Many had access to tools in the EHR that let them organize information about a single patient. They could use it for simple tasks. However, they never spent the time to learn how to use the tool or to configure it to meet their needs. R4_3 shared: "*[Configurable screen]'s relatively new. No training offered for it [. . .] I guess 85% of people don't use it correctly or may not know how to use it.*" The display of information for a single patient seemed at odds with nurses' need to divide attention and effort among two patients, or the RT's need to plan visits to all patients receiving breathing support. R6_1 talked about using the system to view the needs of an individual patient. "*It tells you the patient's condition, code status, SOAPP score, etc. You can set it up however you want [. . .].*" In response, R6_2 asked, "*Wait. Where do you look at it?*" before sharing, "*[...] Oh... yeah. . . I don't use it.*"

4.2.2 ***View Their Communication as Effective Coordination***. Participants thought of themselves as good at coordination; thus, they did not believe that improving coordination might increase WUB adherence. They viewed the chaotic nature of the ICU with rapid shifts in priorities as the real barrier to coordinating their efforts.

Participants strongly equated communication with effective coordination (Fig. 5). As R3_2 shared when asked about the SAT/SBT schedule for coordination: "*I really do feel like our unit relies so much on verbal communication, and I think we do really well with it if you worked there a long time, you know ...*" A salient theme that emerged was the mutual appreciation between RTs and nurses, with participants highlighting how friendly communication facilitated effective collaboration. This differs from previous literature that examines coordination in terms of a clinician's ability to anticipate each other's actions [14, 18]. This emphasis on explicit interaction, rather than shared anticipation, may explain some of the ongoing coordination challenges.

When talking about coordination, participants spoke about the need to be considerate of one another. They seemed more like people who work in proximity to one another than workers who depend on one another. As N6_2 shared: "*I try to be considerate of the RT's time and how busy they are.*" Because the RTs were responsible for all patients across the unit, nurses rarely expected RTs to provide a precise arrival time for when they might perform SBT. As N6_1 explained: "*I tend to give the RT a little bit more room if I know that other patients are dying ...*" This expected flexibility reflected the reality of dynamically prioritizing patients with the most urgent needs. It also revealed their focus on a lack of empathy as the most likely cause of coordination breakdowns.

Nurses shared that they experienced better coordination when working with familiar RTs. The familiarity between the RTs and nurses, as well as the familiarity with patients being treated, made it easier to predict how the day would unfold. As N8_1 noted: "*We usually get a lot of the same RTs assigned to our unit. So I feel like they know the patients and us. We know our patients, and we work really closely together.*" This familiarity fostered a sense of being part of an effective care team.

Interestingly, participants viewed families visiting loved ones in the ICU as a potential disruption. They shared that family involvement can affect how well the WUB protocol is carried out. Family members would approach them as they coordinated SAT and SBT, and would attempt to ask questions that disrupted the whole process. These situations were viewed as impediments to their work. As R4_2 described: "*You get the family from hell that's breathing down your neck.*" R3_3 explained further: "*Sometimes... families come to patients and try to talk. Actually, this disrupts the whole process...*" Family members were also seen as impacting the prioritization of patient care. R4_2 illustrated: "*If this room has five family members in and I need to SBT that patient, I'm probably going to start in a room with zero family members.*" Both SAT and SBT create discomfort for the patient, who can appear peaceful when sedated. The clinicians did not enjoy being seen, adding to a patient's level of discomfort. Family members in the room seemed to add an additional barrier to adherence.

Participants talked about "*pickle phones*," old mobile phones the hospital provided to support coordination. Communication via the

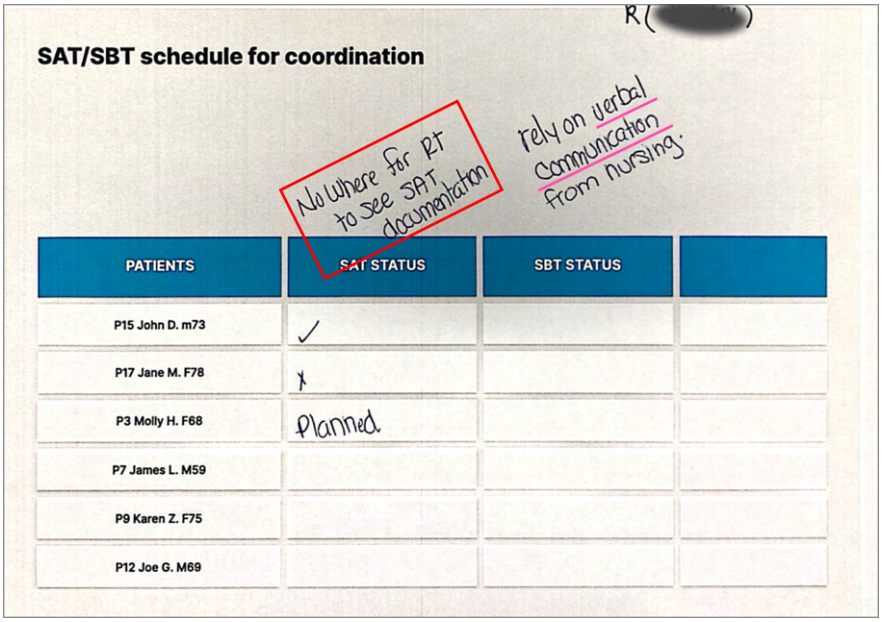


Figure 5: Example of an artifact produced by participants. Although the activity focused on the SAT/SBT schedule, they consistently returned to the verbal communication and documentation problem.

phones disconnected the work from the EHR, which held all the information about each patient. As N6_1 described how they would use the phone to coordinate WUB. "*We call because, like, if I'm, like, if I'm in the room turning the sedation off, sometimes I might just call the RT at that point. So everyone carries a little phone.*" Note: the call happens after SAT is started. They were not calling the RT to find the best time to start the SAT.

Processing data from the first several workshops made it clear that participants equated good coordination to good communication and bad coordination to a lack of empathy. We wanted to disrupt this perspective. To push them to think more broadly about future coordination, we offered some examples where it seemed likely they had witnessed excellent coordination. We talked to them about Starbucks and the coordination between the cashier taking orders and the barista preparing drinks. We probed them by sharing: "*Starbucks has a very highly collaborative process. When a customer orders an iced Americano, the cashier inputs the order and takes payment. The barista receives a sticker with the order and makes the drink. Everything the barista needs is well organized so they can rapidly make exactly the drink the customer ordered. These processes unfold seamlessly without extra conversation.*" They acknowledged the effectiveness in how Starbucks had designed the roles and workplace. They recognized how different this was from their own work environment. N7_2 remarked, "*Yeah. Yeah. Good example. That's good coordination. Thanks for sharing this.*" Starbucks' workplace design achieves smooth coordination without communication. Participants could not imagine how they might achieve a Starbucks-level of coordination within the ICU, a place without real-time data about each patient and without a holistic view of the work each clinician needed to perform.

4.2.3 ***Desire Relief from Documentation Burden.*** Participants sometimes redirected workshops away from WUB adherence and RT-Nurse collaboration with suggestions for where new computing systems might improve their lives. Documentation was typically described as a burdensome issue. They expressed frustration about the volume of required charting, and expressed frustration because they felt the majority of the data they were required to enter did not assist with patient treatment. Documentation was needed for credit even when the data is not needed for patient care. As N6_2 shared: "*When we do a sedation interruption, we need to document everything for us to get credit for it. Both stop the interruption as well as start the interruption, or else we don't get credit for doing a sedation interruption.*" R6_1 nodded and added: "*So we wanted to document it correctly and everything so we get credit for it…*" This is documentation functioning exactly as an accountability regime rather than as a tool for the work [15]: the record exists so the labor can be counted.

The deeper problem was that staff rarely had time to document, and saw inefficiencies with their current documentation practices. As N1_1 explained: "*Our workflow is really old school. We just use pen and paper.*" This practice was sometimes shared with hesitation, suggesting a sense of embarrassment. Others described informal workarounds to save time. As R4_3 summarized: "*Documenting is painful. I think the work is 30% actual work and 70% is charting.*" They never felt they were given enough time to do this important work. They documented their actions long after they were taken. For instance, R1_2 reported: "*I took a picture of the [ventilator] screen for documentation. I think this is saving time and grabbing a static number.*"

Although computers were available in each patient room, RTs believed that they were designated for nurses. As a result, RTs often

entered data outside the patient room, making it more difficult to update records. Participants also found accessing EHR data unnecessarily burdensome. For RTs, accessing patient records often meant leaving the patient's room, disrupting their work. As RT2_1 commented: "*It's annoying to keep looking back at old documents… I wish there was a quicker, easier way than just trying to go through all of that.*" More broadly, participants associated documentation as a significant source of burnout. The perceived imbalance between hands-on care and paperwork was a recurring theme. These reflections suggest that while documentation is essential, current systems fail to support it in ways that enable coordination or support individual work. Instead, they create friction, disrupt shared understanding, and contribute to burnout.

Our ongoing analysis as we ran workshops surfaced concerns about documentation. In response, we shifted the focus of the final two workshops away from dashboards meant to address collaboration and adherence and towards how technology might address their documentation concerns. Participants acknowledged that documentation can be important and valuable for their work. When we probed them to envision a better documentation system, they shared ideas to make it more useful. For example, nurses shared that it was difficult to recall what they had done for each patient. They recognized that having a detailed, real-time log of their actions might improve their collective decision-making with physicians. Similarly, RTs emphasized the importance of viewing all of the changes made to ventilator settings. This helped them recognize important trends in a patient's breathing support.

Participants expressed a strong need for real-time data in the EHR to improve their workflow and patient care, but they noted that current systems offer no way of doing this. Although they emphasized the importance of charting in real time, they also acknowledged that heavy workloads made this unrealistic. N7_2 said: "*Sometimes the most frustrating part about being a nurse is that [I] would rather just do the work... like, you could have a super busy day, and then at the end of that day you're like... I have to chart everything that happened.*" Others criticized inefficiencies in the current system, such as repeated entries, as N7_1 shared: "*There's also a lot of double documentation ... But why are we doing this?*" Several participants reported using their breaks and meal times to update records. As N7_1 shared: "*Documentation should all go real-time. But with a heavier workload with patients, we document during breakfast and lunch time.*" Others described how documentation regularly extended their working hours. N7_2 nodded and added: "*When we work 12 hours, we work 13 plus hours because of documentation.*" RTs noted how nurses often had to delay charting when patient care was priority.

Participants also stated that real-time charting would be nearly impossible during acute care, and particularly during emergencies. They documented on paper entering data into the EHR later when they had time. This delay could negatively impact clinical decision making. As R3_1 shared, "*If nurses don't document SAT in real-time, we might miss the window for SBT. This is why we are shouting at each other.*" They spoke about the benefits that might come from real-time information, something their current practice would never achieve.

To explore this action space, we prompted them to imagine what their work might be like if they had a system that automatically generated charts (Fig 6). This probe sparked great enthusiasm. They immediately recognized that this kind of system could produce useful real-time data. They suggested that it follow the structure of nurses' charting and capture significant changes in a patient's conditions. They saw this as a way to document actions towards standards of care. N7_2 shared: "*Passing on a report to your next person, at least as a nurse, we physically write everything down ... all is handwritten, so this is personally what I do… I have every system...body system [the structure of documentation] written. Yeah... but [it] would be nice if there were some kind of simple digital report [for passing on the report to the next nurse].*" Participants also expanded on and shared their ideas. As R8_2 suggested: "*Oh, I think self-regulating ventilation(automatically adjusts the intake and exhaust of fresh air based on environmental conditions) combined with auto charts may be helpful to RTs.*" And R8_1 noted: "*I think technology would be helpful for documenting when to do things when I'm too busy doing patient care and can't chart until the end of the day.*"

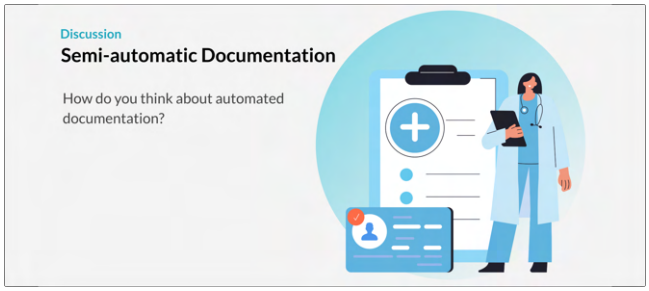


Figure 6: Slide used to prompt discussion during the workshop.

In the last workshop, the conversation around auto-documentation quickly devolved as participants considered the details of a system's behavior. They suddenly began to imagine the system sending alerts. It quickly turned into a dystopian turn-by-turn system where the computer was in charge and they simply worked as the computer's hands. N8_1 commented: "*Oh, document automatically ... when SAT is done, ping RTs.*" This led to concerns about alerts and interruptions. N8_2 responded: "*But I guess it would depend on how respiratory feels about getting notifications. I feel like, rather than autonomy, verbal communication helps the most.*"

At this point, the tone of the discussion shifted, becoming much more negative. Participants quickly turned their attention to the *alert fatigue*. They were wary of documentation tools that might generate additional notifications, associating them with existing ICU systems known for frequent and irrelevant alerts. Their resistance reflected a broader concern about being presented with yet another source of information they felt obligated to monitor. This strong negative perspective made it difficult for them to envision new technology solutions without recalling past frustrations.

4.3 DE1 Reflection: Shift towards Reducing Documentation Burnout

The co-design revealed surprising findings that did not align with insights from prior work on reducing ambiguity and increasing coordination. Rather than embracing the dashboard concept, nurses and RTs questioned a dashboard would change their behavior. This led us to recognize that the dashboard concept framed primarily around adherence and patient outcomes. In effect, it asked workers to change how they work to benefit someone else, with no benefit for themselves. Clinicians perceived the dashboard idea as increasing their workload without improving their own work experience. As a result, clinicians are unlikely to change the way they work unless they receive some benefit from the change.

In Design Experiment 1, we came to see that improving coordination around WUB requires addressing clinician burnout as a starting point. This raised a central question for our work regarding whether we can design a computing system that improves clinician's work experience while also improving patient outcomes. To further explore how computing might be accepted by clinicians and increase adherence to standards of care, we broadened our focus beyond the dashboard and searched for more holistic solutions that might transform clinical practices around WUB.

5 Design Experiment 2: Sketching and Enacting the Design Opportunities

5.1 DE2 Process

We aimed to envision and assess concepts that simultaneously addressed adherence and documentation burnout. This involved two distinctly different activities: (1) sketching concepts and (2) co-design sessions where participants enacted and reflected on future work practices.

The research team collaboratively brainstormed and sketched several new concepts based on insights from DE1. We synthesized the key needs and challenges expressed by clinicians and the WUB literature. We then mapped these to potential design solutions. Our iterative process resulted in a final concept that automatically documented actions related to WUB and made the status of WUB activities known across the ICU. We then detailed how the concept should interact with clinicians and the value it might offer.

To assess this concept, we conducted another round of co-design sessions, drawing on Brandt's idea of an enact co-design activity, where participants act out a possible future in order to reflect on the future that could and should be [17]. To help participants better channel their authentic feelings of work, we worked with a healthcare simulation and education center. This training facility simulates clinical practice, and has hospital rooms, ICU equipment including ventilators, and interactive mannequins placed in bed to simulate patients being cared for (Fig. 7).

We recruited 3 nurses and 3 RTs (N=6) who performed WUB. We held three sessions summarized in Table 2. All had at least 1 nurse and 1 RT except session 2, where only one of the three recruited participants attended. Participants were compensated $75 for their time.

The sessions ran 60 to 90 minutes. We asked participants to recall a recent time they performed either SAT or SBT and then to recreate that performance in our simulation room. We asked them to deviate from the previous performance by vocalizing their action, something the auto-documentation system might need to capture their actions. Following their performance, we held a short discussion, probing participants on what such a system might be like and how it might impact their work. We video recorded the sessions, produced transcripts with Otter.AI, and wrote field notes.

Table 2: Participant Demographics for DE2's Co-design Sessions

Participant Code	Role	Gender	Years of Experience
N1	Nurse	F	6
RT1	RT	F	28
RT2	RT	M	10
NT	Nurse	F	38
N3	Nurse	F	3
RT3	RT	F	6

We analyzed the video data, transcripts, and field notes the same way as we used in DE1, by creating affinity diagrams [35] using the collaboration software, Figma. We individually reviewed the data and had meetings to cluster notes and to discuss and refine emerging clusters, drawing out key insights.

5.2 DE2 Findings

DE2 produced a high-level auto-documentation concept that captures WUB-related actions in real-time and make their status visible at the ICU level.

5.2.1 ***Sketching an Auto-documentation System.*** Sketching sessions produced several potential design solutions:

- ***Log Clinician Actions Based on Standards of Care.*** Capture actions and coordination required by standards of care, such as nurse–RT coordination of SAT and SBT. These data could support adherence monitoring, accountability, process improvement, and clinical AI development.
- ***Improve Documentation Tools.*** Integrate documentation into clinical workflows to capture communication, activities, and device data in real time. This could reduce delayed charting and after-hours work while improving data quality and clinicians' work experience.
- ***Dynamic Staffing.*** Predict ICU workloads to align staffing with patient needs. Although this could improve staffing responsiveness, it could also increase reliance on contingent employment or lead to staffing reductions.
- ***Predict Family Visits.*** Anticipate visit times to help schedule SAT and SBT, which clinicians sometimes avoided during family visits. Conversational agents could answer family questions and gather visit plans, but this approach could raise serious privacy concerns.

Through iterative sketching and critique, we speculated the auto-documentation concept that best addresses both WUB adherence and documentation burden simultaneously. We envisioned an auto-documentation system that monitors and logs clinicians' actions



Figure 7: Study environment of DE2's Co-Design Sessions.

related to the WUB standard of care. Unlike digital scribes that capture everything that is said between a patient and clinicians, our concept limits what it captures to actions and audio related to exclusively to WUB. This concept involves adding sensors in the patient room (cameras, depth cameras, microphones). The concept encourages clinicians to vocalize their actions. This is especially important for capturing their judgments, such as assessing if a patient qualifies for SAT and SBT. When they disqualify a patient, vocalizing their concern would allow the system to make note of why they disqualified this patient at this time.

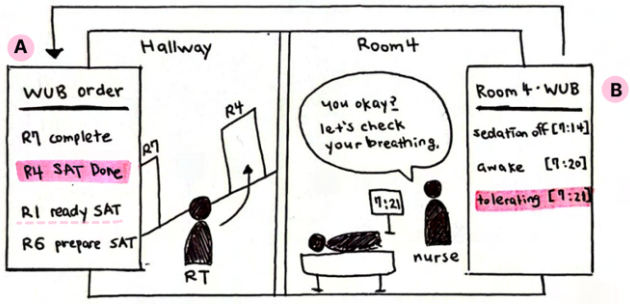


Figure 8: Two touchpoints of the auto-documentation concept. (A) A hallway status display lists patients in the order WUB will be performed, room where WUB is underway is highlighted. (B) A bedside display in the patient's room gives the nurse feedback on the actions it has recognized and confirmed events update the EHR and the hallway dashboard as the work happens.

The concept involves hallway status display (Fig. 8, A). It gets the RT to a patient's room while that patient is awake (done with SAT). It displays a ranked list of patient rooms for all patients who qualify for WUB, along with each room's status (WUB completed, SAT in progress, SAT completed, SBT active, WUB pending), and marks where the RT is working now and which room is next. Nurses could see where the RT was on the list and estimate when to start SAT. In the patient's room (Fig. 8, B), the bedside display gives clinicians feedback on the WUB actions it has recognized. When clinicians initiate the WUB protocol, the system listens and records the actions related to the WUB. The data include documentation and timestamps recorded in the EHR system and transferred to the hallway dashboard in real-time.

The concept was intended to make WUB actions visible. By capturing real-time data, the system could support nurses and RTs in coordinating SAT and SBT while reducing the documentation burden. Whether nurses and RTs would accept such a system remained an open question, the following co-design was designed to answer it.

5.2.2 ***Co-design Sessions: Enacting and Assessing a Possible Future.*** When we initially explained the concept of the auto-documentation system, participants appeared pleased. Interestingly, several assumed such tools would only be for doctors. They struggled to imagine that an IT system would make their work easier. Several had observed physicians using digital scribes, and they wished they had something similar. Once we clarified that we are designing for ICU nurses and RTs and for care coordination, they welcomed the concept and expressed a strong desire to adopt it to reduce their documentation workload.

When first acting out a recent performance of WUB with the vocalizations needed for the conceptual system, several participants explained their actions to the facilitator instead of vocalizing. We addressed this by instructing them to pretend the research team was not in the room and to focus on the interaction itself. This new instruction allowed participants to more comfortably engage in natural role-play, interacting solely with the mannequin and conceptual auto-documentation system.

In reflecting on their current work practices and the potential impact of auto-documentation, participants expressed frustration with their current state. They viewed much of current documentation to be a waste of time that could otherwise be spent caring directly for patients. Thus, they were willing to adopt a system that reduced this load, even if it required audio or video recording. Specifically, participants appreciated the system as a valuable time-saving tool in which they can spend less time on documentation and get real-time data in the ICU. Some clinicians even expressed

strong support and a desire to implement an auto-documentation system in their ICU workflow. Rather than perceiving continuous recording as a privacy concern, participants felt that being recorded enhanced their professional accountability. RT2 mentioned "*I would like to see more surveillance. At work in an ICU, privacy and lack of accountability is a problem, and I think the lack of accountability, even with our technology, still exists and persists. So if surveillance can mean accountability, that's great. I think we need that (being recorded and automatically documented).*"

Once they began vocalizing, most of the participants seemed comfortable with the behavior. Several shared that they regularly vocalize their actions to reduce patient anxiety. Enacting the behavior made it clear that vocalizing was much less effort than documenting. They shared that they regularly speak to patients about less observable outcomes, such as whether a procedure succeeded or failed. N2 shared, "*I found I was always doing it (vocalize my work).... (telling patient) you doing good. You're a little bit anxious and sleepy. We'll try (SAT) again later. I'll try secretions, so we get a better handle on them.*" The enactment activity was also valuable in capturing how participants preferred to interact with the system, highlighting their desire for real-time feedback on its interpretations. RT1 noted "*but you pull it over, [...] (I still want to review) review everything before it's pulled over. [...] I'm the one whose name gets put.*" N2 also shared, "*I might [be] supposed to check it [the auto-documented chart] just to make sure that it was accurate.*" When the system reduced their documentation workload, participants wanted a screen where they could confirm the system's interpretation and the generated document.

Participants also highlighted the specific tasks they wish to be auto-documented, which included high-workload tasks such as emergency situations, WUB protocols, and complex skin assessments. These tasks typically require extensive narrative documentation, including details of the entire event, which clinicians were involved, and the patient's condition and the assessments. As participants explained the heavy documentation burden of their work, they naturally envisioned how the system might capture their work. Participants noted that ideally they are expected to document every two hours (which is impossible), and that it will be useful when the system generates document in real-time and they can confirm every two hours. N1 shared "*It would be nice if there's an automatic every two hours, maybe that it pulls over your vital signs more easily.*"

At this point, we could reveal that clinicians would accept the concept and how clinicians think about their behavior change with vocalizing for the system. Beyond this, we could explore how clinicians envision and desire to interact with an auto-documentation concept.

5.3 DE2 Reflection

In our sketching and brainstorming activity, a preferred future emerged that simultaneously addresses adherence and documentation burnout. Reflecting on this transition, we came to view reducing documentation workload as a starting point for connecting clinicians' needs with our broader goals of care coordination and adherence. This led us to envision clear values of the auto-documentation system and how it should work. Clinicians' enactments revealed a strong desire to integrate the system into their workflow and how they want to interact with the system. DE2 allowed us to assess how the system can be accepted by clinicians and its impact.

Translating multimodal inputs into clinical text is not a technical innovation [16, 70]. Commercial ambient scribes capture the patient–physician consultation and aim to bill and complete the notes [1, 32, 75]; the clinician's benefit is a shorter evening of charting, and the artifact produced is a document. Our concept differs in what it is for. It captures a protocol rather than a conversation, it captures care coordination rather than patient-physician communication, and its output is a real-time signal that coordinates the next clinician's action, not a note filed after the fact. That difference is what makes the same sensing technology into care coordination.

5.3.1 ***Limitation on DE2.*** Our study has two primary limitations. First, DE2's co-design session was limited by its small sample size. Although we intended to explore surface design opportunities as a generative design instead of establishing generalizable adoption [31, 100], we have a small sample size. Second, we co-designed exclusively with nurses and RTs, even though the concept introduces cameras and microphones into rooms occupied by patients and their families. Neither stakeholder group was represented in this initial design process. We aim to address these limitations in future work.

6 Discussion and Open Research Questions

We conclude by reflecting on the overall design process within a high-stakes, dynamic environment, and outlining open questions for future research. Clinical standards help to ensure the practice of evidence-based medicine [46]. Unfortunately, implementing these standards can be challenging, especially in contexts like the ICU, where work involves life and death decision-making [7, 22]. Clinicians may not be able to adhere to clinical standards. Standards that require collaboration seem especially challenging, given the lack of real-time data related to patient treatment, as well as a lack of tools to help coordinate the work of all clinicians working on a ward, within a department, or within a single ICU.

6.1 Designing the Right Thing in High-Stake Environments

In high-risk domains such as the ICU, traditional, linear approaches to user-centered technology design are particularly prone to failure. Our findings from Design Experiment 1 illustrate this challenge: participants did not embrace dashboard-based solutions, despite their advocacy in prior research. Rather than viewing this rejection as a failed design attempt, we conceptualize this reframing as "*Drifting by Intention* [45]."

To bridge the gap between design research and real-world deployment, HCI researchers and designers must look beyond whether a system is usable or technically effective. Instead, we must critically consider how the proposed technology can shape the work of practitioners, including their daily practices, responsibilities, coordination, and existing work practices [29]. Design Experiment 2 demonstrates this point. By focusing on reducing documentation burnout to improve adherence and create tangible value for clinicians, rather than simply introducing new technology, we found we could generate benefits for a broader range of stakeholders across the ICU.

6.2 Auto-documentation Beyond the ICU

Our Design Experiments shifted our thinking away from shared dashboards to support collaboration and towards a concept that automatically documents clinical tasks related to a standard of care. We speculate an auto-documentation could potentially improve adherence, improve working conditions for clinicians, and produce better quality healthcare data. We anticipate that our design could do more than help with WUB. It has the potential to automatically document all the standards of care for individual patients. Further, we anticipate that our envisioned system can address similar challenges in settings beyond the ICU. The system could provide value for sequential clinical tasks requiring time-sensitive coordination across diverse clinician roles. This is particularly relevant for workflows governed by detailed standards of care that demand heavy narrative documentation. For example, we can imagine an auto-documentation system like this helping in the emergency department, which also has detailed procedures and requires coordination.

6.3 Open Research Questions

Prior literature proposed a coordinated WUB workflow to enhance care coordination and adherence among nurses and RTs (Fig. 1) [99]. Future research can discover an AI WUB workflow that fosters collaboration between clinicians and the auto-documentation system to further improve WUB adherence. This reveals several HCI research questions, including: How should the system provide feedback while clinicians work, and how does this feedback impact interactions with patients, families, and colleagues? How will healthcare administrators react to and adopt such tools? What level of technical capability [27, 58] is enough for frontline clinicians to find an AI system useful and satisfying to use? How can we evaluate technical feasibility to measure system accuracy and examine how clinicians repair AI inference errors?

These questions touch on HCI issues around Human-AI interaction. They also spread beyond the typical boundaries of HCI research by touching on governance and regulation and on the financial operation of the healthcare enterprise. If HCI wishes to move forward with these innovations, researchers will need to bring on new collaborators and subject matter experts who can bring knowledge about these missing areas of concern, helping HCI research translate into clinical practice.

7 Conclusion

Through two design experiments with ICU nurses and RTs, we explored how technology might support adherence to the WUB. In DE1, participants questioned whether a shared dashboard would improve coordination and emphasized documentation burden. This prompted us to explore how addressing documentation needs might also support coordination. In DE2, the research team sketched an auto-documentation concept intended to reduce documentation workload and generate real-time WUB data. Clinicians' enactments extended the concept, surfacing willingness to vocalize actions and desire for how they would like to interact with the system. We contribute a situated account of Drifting by Intention in ICU co-design and a design direction connecting documentation support with care coordination. Future work should examine the concept's feasibility and effects on workload, coordination, and adherence.

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